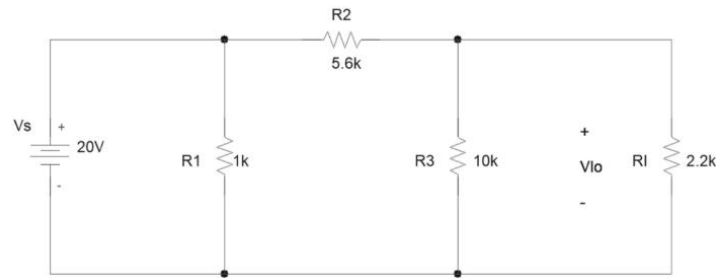
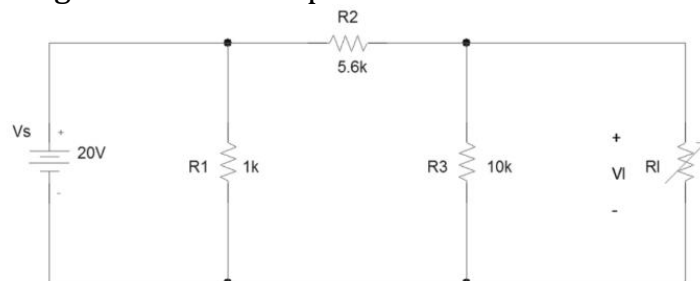


Report Task (Experiment 8)

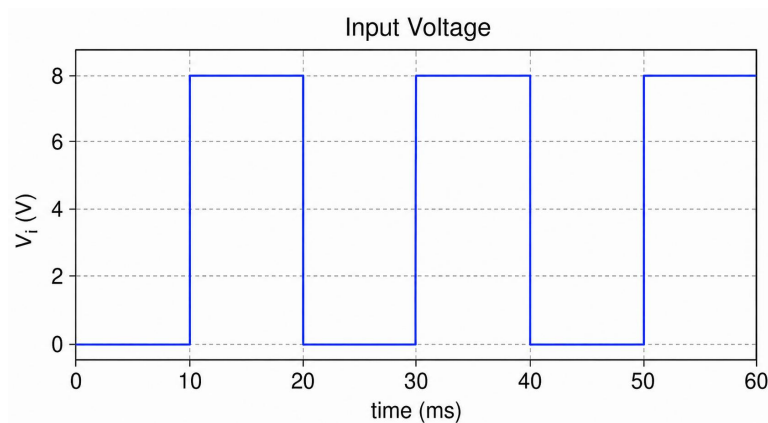
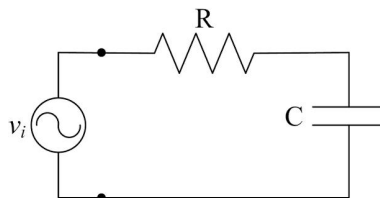
- 1) Verify **Thevenin's Theorem** for the following circuit:



- 2) For the circuit shown below, determine the value of R_L for which the power delivered to the load is maximum. Also, calculate the corresponding maximum load power.



- 3) For the RC circuit shown below, plot the capacitor-voltage waveform $v_c(t)$ for the given square-wave input voltage under each of the following conditions:



- i. $R = 2 \text{ k}\Omega$ and $C = 1 \text{ }\mu\text{F}$
- ii. $R = 2 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$
- iii. $R = 2 \text{ k}\Omega$ and $C = 0.1 \text{ }\mu\text{F}$

Compare the three waveforms and briefly explain the effect of the RC time constant on the capacitor's charging and discharging response.